

Date	1 July 2026
Team ID	-
Project Name	Book-Store
Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II: SOLUTION REQUIREMENTS

Functional Requirements

Following are the functional requirements of the proposed Book Store solution. This table details each Functional Epic alongside its associated Sub Requirements and user stories.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	• Customer registration and login using form fields. • Seller signup with pending status awaiting approval. • Secure system administrator session access gateway.
FR-2	Catalog Navigation	• Searching books by title or author. • Filtering listed items by category tabs (Fiction, Science). • Detail book view showing price, author, and description.
FR-3	Wishlist Controls	• Add books to customer local wishlist array. • View and delete saved books from customer wishlist tab.
FR-4	Checkout & Orders	• Instant "Buy Now" checkout form collecting name/address. • Automated database order generation and stock deduction. • Status update updates for sellers (pending, ontheway, delivered).
FR-5	Admin Control Panel	• Toggle seller "isApproved" state. • Edit/delete user details and delete customer orders.

Non-Functional Requirements

Following are the non-functional requirements of the proposed Book Store solution. This details system attributes such as security, usability, availability, and response metrics.

FR No.	Non-Functional Requirement Description	
NFR-1	Usability	• Highly responsive user interface scaling dynamically using Bootstrap. • Clean cream/beige palette (#FDFBF7) optimized for high readability.
NFR-2	Security	• Passwords encrypted using Bcryptjs (10 salt rounds). • Route validation protected via JWT tokens in HTTP Authorization headers.
NFR-3	Reliability	• Strict Mongoose schemas preventing inconsistent data fields. • Elegant database error logs to isolate server connection issues.
NFR-4	Performance	• Search index queries on MongoDB Atlas optimized for speed (under 120ms). • Fast page load times (under 2 seconds) by caching static assets.
NFR-5	Availability	• Hosted on Render paired with MongoDB Atlas globally distributed clusters. • Standard service uptimes of 99.9% or higher.
NFR-6	Scalability	• Decoupled MVC routers making addition of future features or models easy. • Horizontal cluster scaling via MongoDB Atlas.